GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Sampling Techniques Activity

Definitions, applications and examples of sampling methods relevant to the research problem

Student Name	Kothari Tarj Mehulbhai	Enrollment No.	260205020001
Subject	Research Methodology (METXX11501)	Semester	1 (M.Tech)
Activity	ALA 3	Marks	10

Researchproblemusedintheseactivities: ComparativeSearchEfficiencyofaRandomizedSkipList,a DeterministicRed-BlackTreeandaSelf-AdjustingSplayTreeunderDifferentInsertionOrdersandAccess Patterns.
(Illustrativeproblemselectedforthissubmission;replaceitwiththefaculty-assignedproblemifadifferentoneis given.)

1. Aim

To prepare a report on various sampling methods, including **definitions, applications and examples relevant to the research**. In addition to describing each method, the report **demonstrates** the probability samplingmethodsonarealdatasetgeneratedfortheresearch(searchcostsof100,000keys)andshows how sample size is determined.

2. Fundamentals of Sampling

Term	Definition	In this research
Population (universe)	Thecompletesetofunitsaboutwhich conclusions are wanted.	All 100,000 keys stored in one skip list (demonstration);allpossiblerandomdatasets (experiment);allstudents/developers(planned survey).
Sampling frame	The list from which the sample is drawn.	Thesortedlistofkeys;therolllistofthe student population.
Sample	A subset of the population actually studied.	500keys;30replications;asetofsurvey respondents.
Parameter / statistic	Apopulationvalue/thesamevalue computed from the sample.	Population mean search cost μ= 32.657 nodes(census);samplemeanistheestimate.
Sampling error	Difference between a sample statistic and thetrueparametercausedbystudyingonly part of the population.	Standard error of the estimate (Section 5).
Bias	Systematicerrorcausedbyapoorselection method.	Conveniencesampleofthesmallestkeys (Section 5).

Need for sampling

Studying every unit (a census) is often too costly, too slow or impossible – for example, surveying all software developers, or testing every possible input of an algorithm. A well-chosen sample gives reliable estimates at a fraction of the effort, and the precision can be quantified.

Principles of good sampling

- **Representativeness:** the sample should mirror the population on the characteristics that matter.
- **Randomness:** every unit has a known, non-zero chance of selection (for probability sampling), which removes selection bias.
- **Adequacy:** the sample size must be large enough for the required precision and power (Section 6).
- **Precisionandefficiency:**usedesigns(stratification,systematicselection)thatreducevarianceforthe same cost.
- **Practicability:**acompleteframemustbeavailableandselectionmustbefeasiblewithintimeand budget.

3. Types of Sampling Techniques

Probability (random) sampling

Technique	Definition	Application	Examplerelevanttothis research	Advantages/ limitations
Simple random	Every unit has an equal chance; units drawnatrandom(e.g. random numbers) from the frame.	Homogeneous populationswith a complete frame.	Choose500ofthe100,000 keysatrandomtoestimate average search cost.	Unbiased,simple;needs afulllist;maymisssmall sub-groups.
Systematic	Select every k-th unit after a random start (k = N / n).	Ordered lists, productionlines, files.	Takeevery200thkeyofthe sorted key list starting at a random position.	Easyandspreadevenly; can be biased if the list has a periodic pattern.
Stratified	Dividethepopulation into homogeneous strata and sample from each (proportionalorequal allocation).	Heterogeneous populationwith known sub-groups.	10key-rangestrata(deciles); 50 keys from each. For a survey: strata = year/programme.	Guarantees representation; more precise;requiresstratum information.
Cluster	Divide the population intoclusters(naturally grouped),selectsome clusters at random and study all units in them.	Geographically spreadorlarge populations without a complete list.	Choose 5 blocks of 100 consecutive keys. For a survey:choose5classesand study all students in them.	Cheapandpractical; usually less precise (unitsinaclusterare alike).
Multistage	Sampling in stages (e.g. institutes →classes→students).	Very large, dispersed populations.	Select colleges, then departments,thenstudents for a state-wide survey.	Flexible; error accumulatesateach stage.

Non-probability sampling

Technique	Definition	Application	Examplerelevanttothis research	Advantages/ limitations
Convenience	Unitschosenbecause they are easy to reach.	Pilotstudies, quick exploratory work.	Using only the first 500 (smallest)keysinthelist; asking classmates in the same lab.	Quickandcheap;high risk of bias; results cannotbegeneralised.
Judgement (purposive)	Researcher selects units believed to be typicalorinformative.	Expertstudies, case studies.	Choosingexpertdevelopers who maintain library data structures for interviews.	Targets relevant cases; depends on the researcher'sjudgement.
Quota	Fixed numbers are taken from each category,butselected byconveniencewithin it.	Marketsurveys whennoframe exists.	Interview20first-yearand20 final-year students.	Controls composition; selectionwithinquotais non-random.
Snowball	Initialrespondents refer further respondents.	Hidden or hard-to-reach groups.	Findingdeveloperswhohave implemented skip lists in production by referral.	Reaches rare groups; samplemaybeaclosed network.

4. Where Sampling Occurs in the Research Problem

Stage	Population	Sampling method used
Replications of the experiment	All possible random data sets	30independentrandomseeds(simplerandom sampling of data sets)
Search keys used as queries	All keys stored in the structure	Simple random with replacement (uniform workload);90%fromahotsetof1%ofthe keys (skewed workload, a two-category sample)
Key values stored	Integers 1–999,999	Simple random sample without replacement
Planned survey (ALA 4)	Students/developers (assumed N = 600)	Stratified random sampling by year of study; samplesizefromCochran'sformula(Section 6)

5. Demonstration: Comparing Sampling Techniques on Real Data

Population: a skip list holding N = 100,000 random keys. The value for each key is the number of nodes inspected to find it (a census was run: mean μ= **32.657**, standard deviation σ= 4.871, range 17–59). **Task:** estimate $\mu$from a sample of n = 500 keys. Each technique was repeated 2000 times and the 2000 estimates were compared with the true value.

Sampling technique	Mean of estimates	Bias	Standard error	RMSE	Designeffectvs simple random
Simple random	32.663	+0.005	0.217	0.217	1.0
Systematic (every 200th key)	32.653	-0.004	0.133	0.133	0.37
Stratified (10 key-range strata)	32.664	+0.007	0.198	0.198	0.83
Cluster (5 blocks of 100 keys)	32.645	-0.012	1.7	1.7	61.22

Bias=meanofestimates−μ.Standarderror=standarddeviationoftherepeatedestimates.Designeffect=(standard error)^2 / (standard error of simple random)^2; values below 1 mean more precision than simple random sampling.

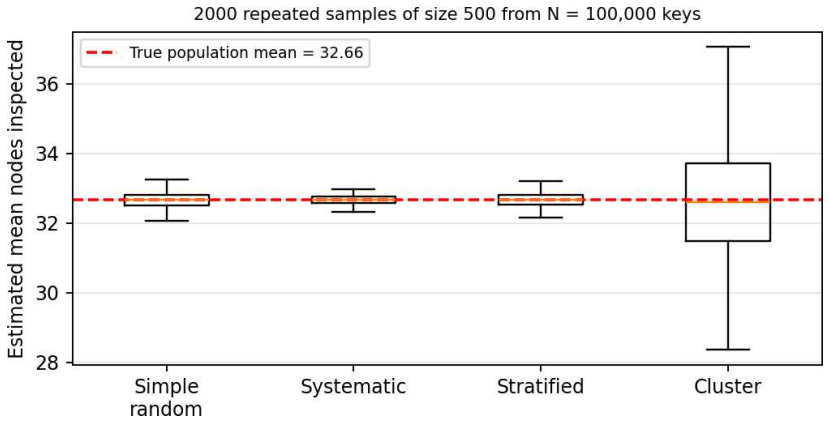


Figure 1: Distribution of the estimates for each probability sampling technique.

Non-probability examples (one sample each, so no standard error)

Technique	Sample	Estimate	True μ	Error
Convenience	The 500 smallest keys (front of the list)	25.766	32.657	-6.89
Judgement (purposive)	The500largestkeys,chosenas'typical heavy users'	35.236	32.657	+2.58

Findings

- **Simple random sampling** was unbiased (bias +0.005) with standard error 0.217 nodes.
- **Systematicsampling**wasalsounbiasedandmoreprecise(designeffect0.37)becausethesampleis spread evenly through the ordered list.
- **Stratified sampling** was unbiased and slightly more precise (design effect 0.83); the gain is small because search cost varies little between key ranges.
- **Cluster sampling** was unbiased but much less precise (design effect 61.22): neighbouring keys have similar search paths, so 500 keys from 5 blocks carry far less information than 500 scattered keys.
- **Convenience sampling** was badly biased (-6.89 nodes, about 21% too low): the smallest keys are cheapertoreach.Itserrorwouldnotshrinkevenwithrepeatedsamplingandcannotbeestimatedfrom the data – a typical problem of non-probability samples.
- **Lesson:** use probability sampling when inference to a population is required; choose the design according to the structure of the population (ordered →systematic; sub-groups →stratified; natural groups →cluster, accepting a precision loss).

6. Determining the Sample Size

Survey sampling: Cochran's formula

Toestimateaproportionwithmarginoferroreataconfidencelevelwithz-valuez(≈1.96for95%):$\mathbf{n_0 = z^2 \cdot p(1-p)/e^2}$.Withthemostcautiousp=0.5ande=0.05,n_0 =384.16→**385**respondents.Forafinite population of size N the correction $\mathbf{n = n_0 / (1 + (n_0 - 1)/N)}$ is applied:

Population size N	Required sample n
200	132
600	235
2,000	323

Population size N	Required sample n
10,000	370

For the planned survey of N = 600 students, **n = 235** would be needed, allocated proportionally to year of study.

Experiments: number of replications (power analysis)

For comparing two groups with a t-test (α= 0.05, power = 0.80), the required number of replications per group depends on the effect size d to be detected:

Effect size d (Cohen)	Interpretation	Replications per group
0.2	small	394
0.5	medium	64
0.8	large	26
1.2	very large	12

The experiment used **30 replications per condition**, which is enough to detect effects of about d ≥0.75 with80%power.Becausetheeffectsofinterest(structuretype,accesspattern)wereexpectedtobelarge, this size was adequate.

7. Sampling and Non-sampling Errors

Type	Cause	Control
Sampling error	Only part of the population is observed.	Largern;betterdesign(stratification);report standard errors and confidence intervals.
Selection bias	Non-randomorincompleteframe (convenience samples).	Probability sampling; complete frame.
Non-response bias	Selected respondents do not answer.	Follow-ups;compareerespondentswiththe frame.
Measurement error	Faulty instrument or wording.	Pilot testing; calibration (ALA 4).

8. Conclusion

Sampling methods were defined, classified and demonstrated. The demonstration with 100,000 keys showed that simple random, systematic and stratified samples all give unbiased estimates of the population mean, that a cluster sample is much less precise for the same number of units, and that a convenience sample can be heavily biased. Sample-size calculations (Cochran's formula for surveys, power analysis for experiments) show how the research design determines the amount of data needed.

References

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